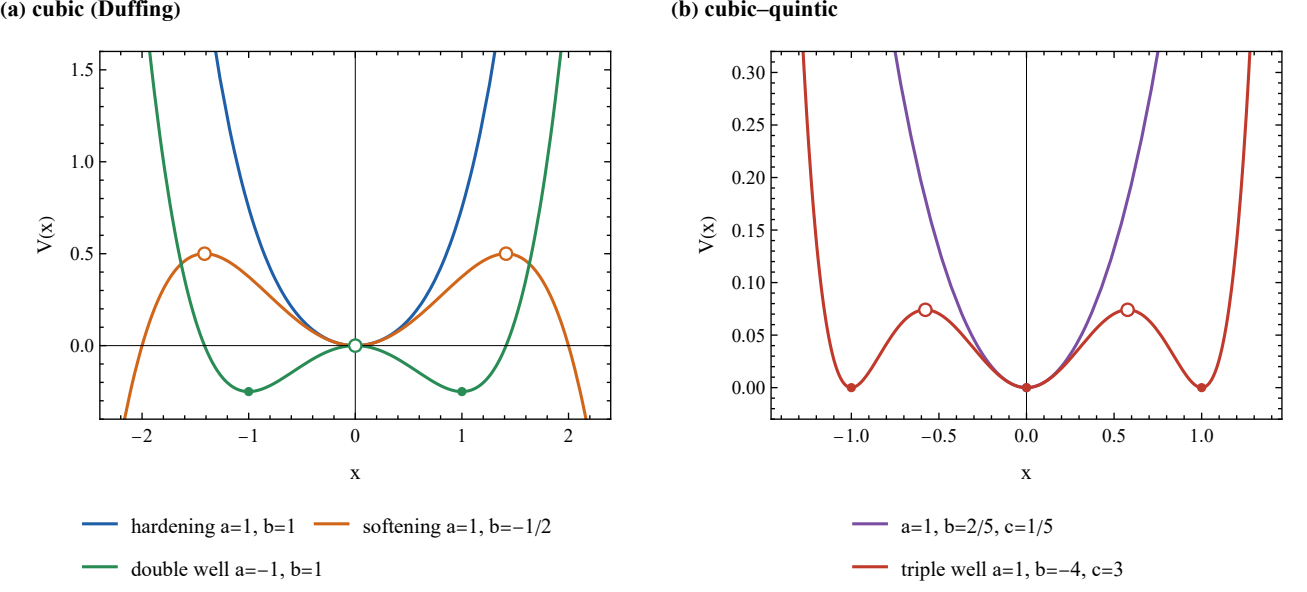

(a) cubic (Duffing)
(b) cubic–quintic
V(x)
x
1.5
1.0
0.5
0.0
−2
−1
0
1
2
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−1.0
−0.5
0.0
0.5
1.0
hardening a=1, b=1
softening a=1, b=−1/2
double well a=−1, b=1
a=1, b=2/5, c=1/5
triple well a=1, b=−4, c=3